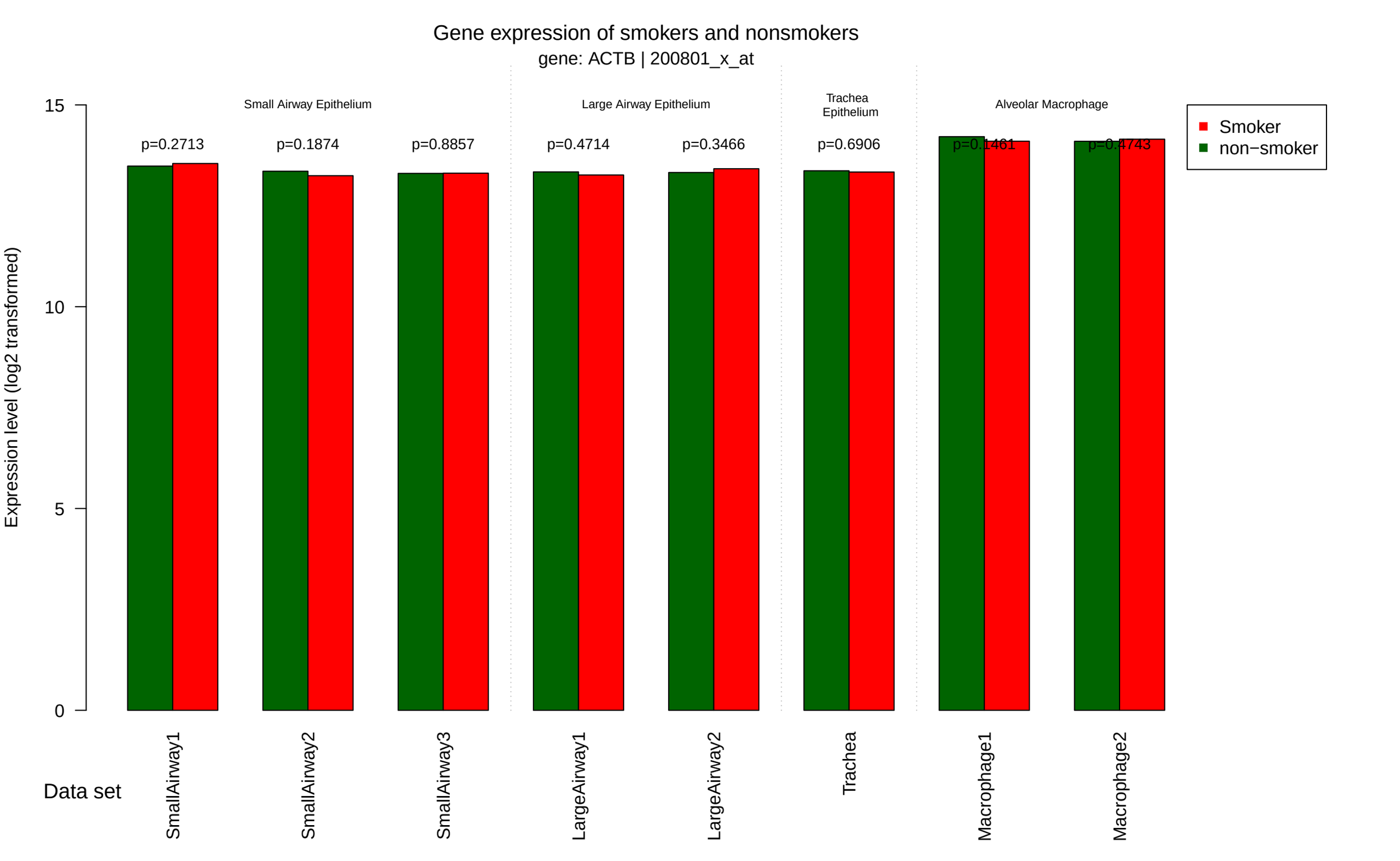
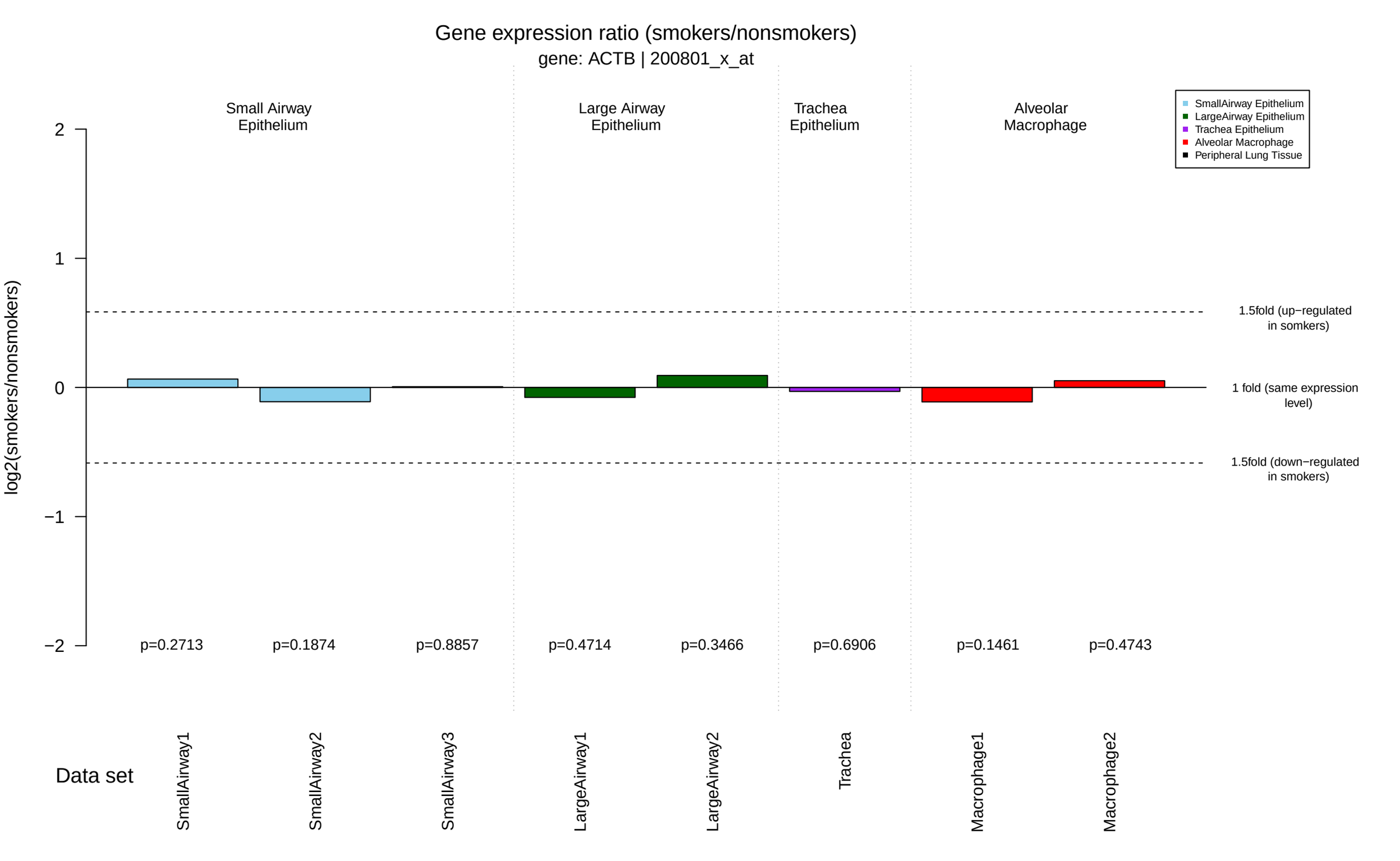
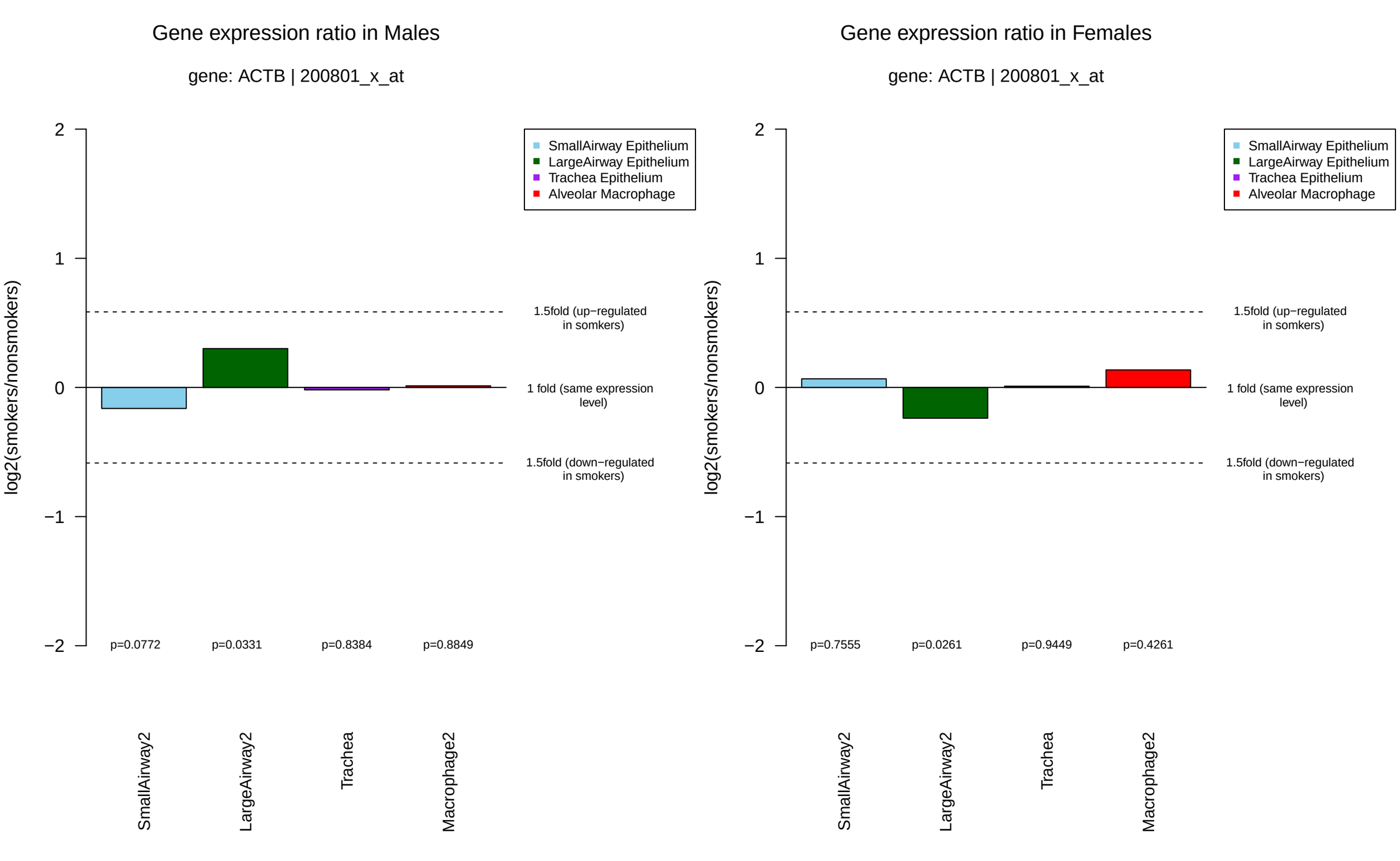
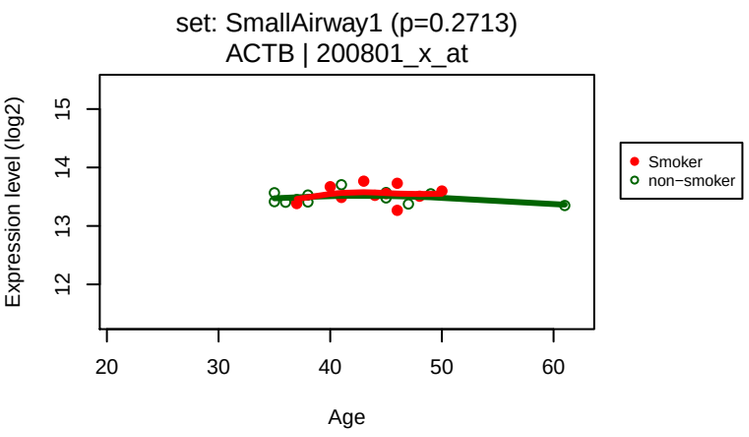
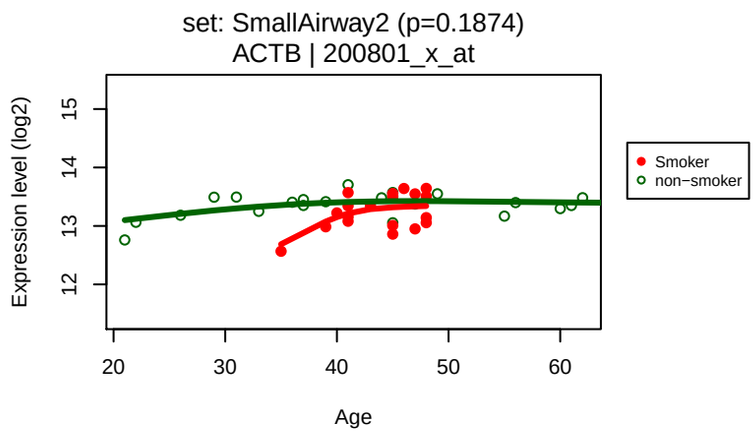
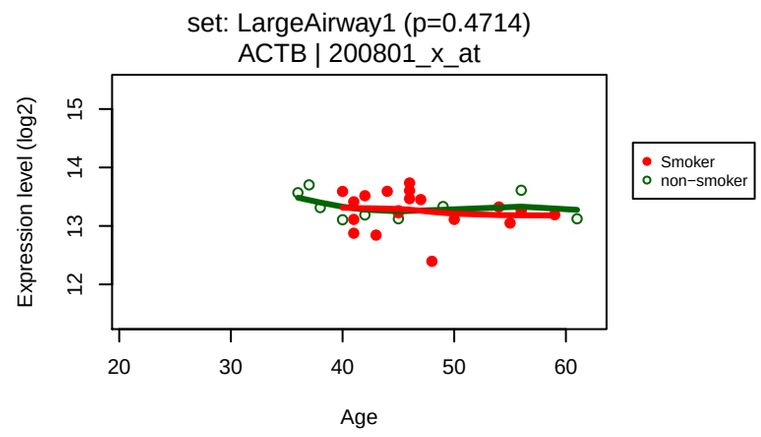

Gene expression of smokers and nonsmokers
gene: ACTB | 200801_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2713
p=0.1874
p=0.8857
p=0.4714
p=0.3466
p=0.6906
p=0.1461
p=0.4743
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

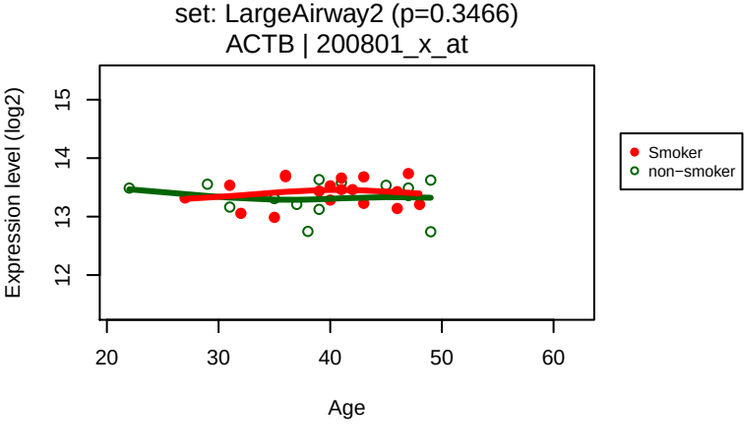

Gene expression ratio (smokers/nonsmokers)
gene: ACTB | 200801_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2713
p=0.1874
p=0.8857
p=0.4714
p=0.3466
p=0.6906
p=0.1461
p=0.4743
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

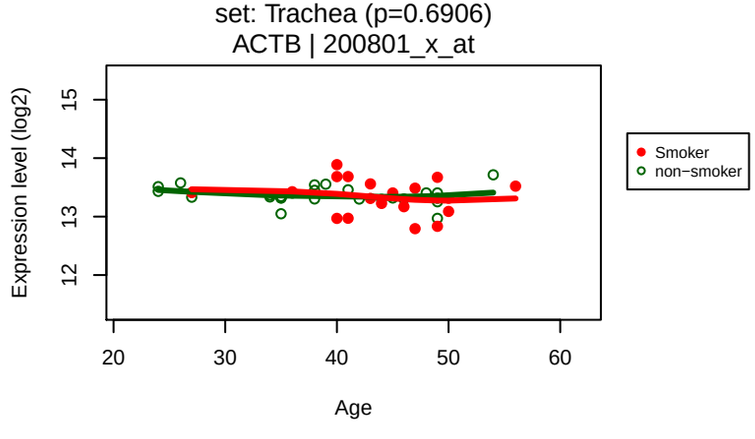

Gene expression ratio in Males
gene: ACTB | 200801_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0772
p=0.0331
p=0.8384
p=0.8849
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTB | 200801_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7555
p=0.0261
p=0.9449
p=0.4261
SmallAirway2
LargeAirway2
Trachea
Macrophage2

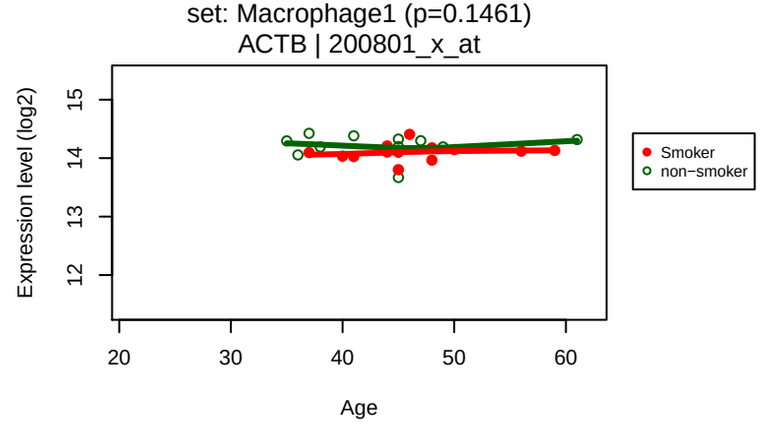

set: SmallAirway1 (p=0.2713)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker

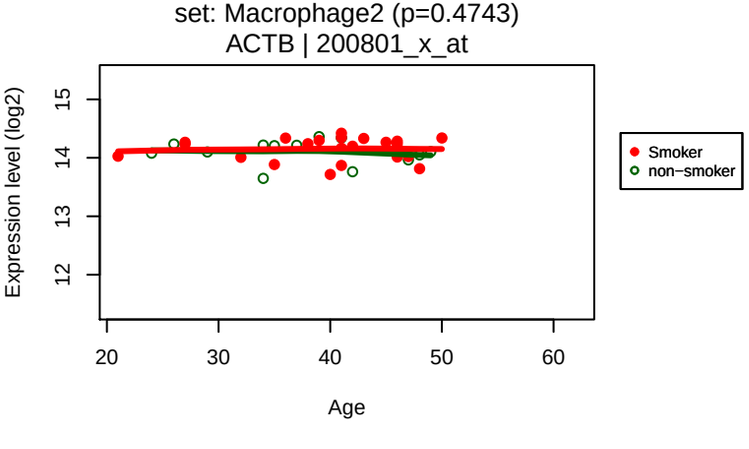

set: SmallAirway2 (p=0.1874)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker

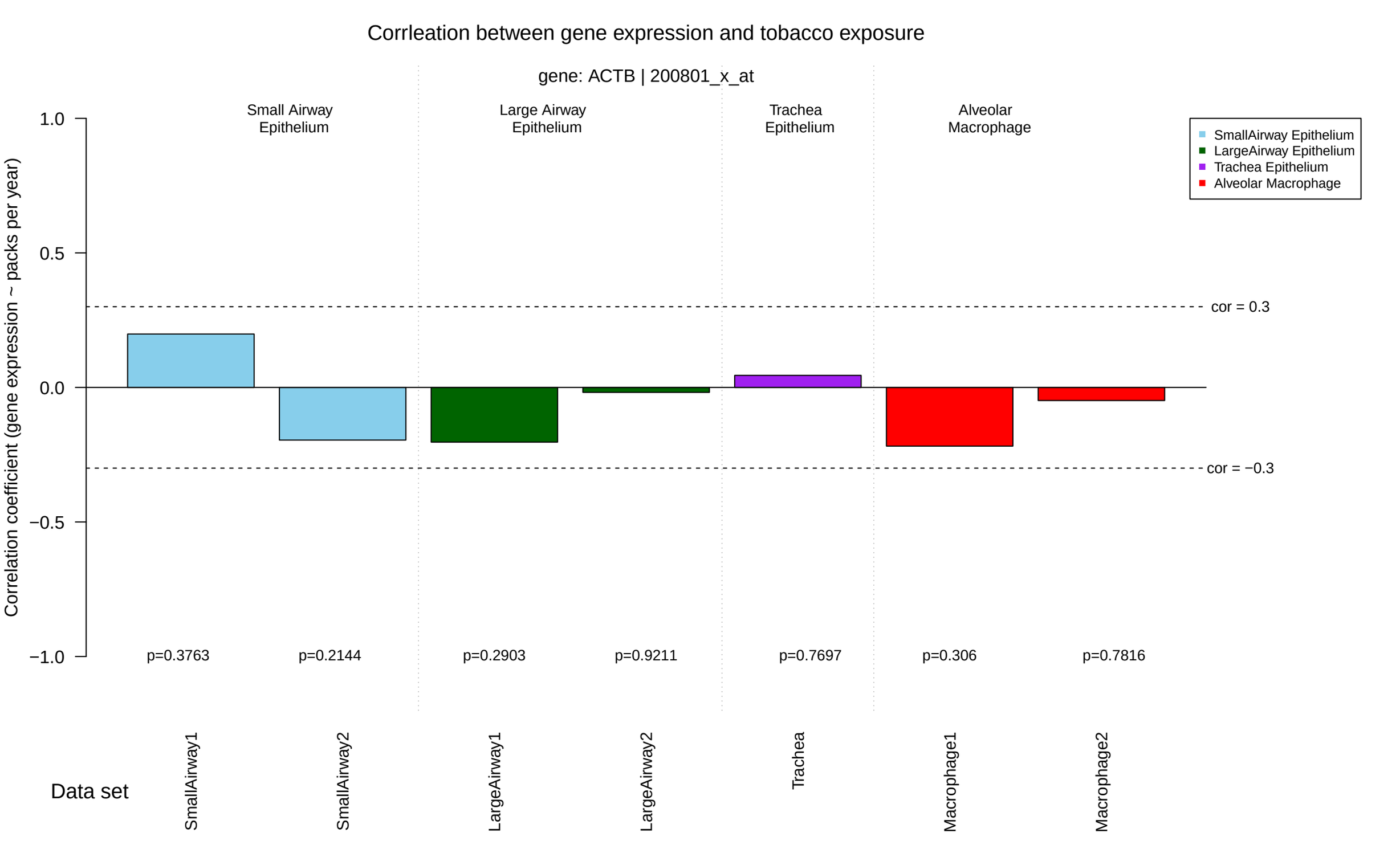

set: LargeAirway1 (p=0.4714)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3466)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6906)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1461)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4743)
ACTB | 200801_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTB | 200801_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3763
p=0.2144
p=0.2903
p=0.9211
p=0.7697
p=0.306
p=0.7816
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2